



João de Jesus Costa

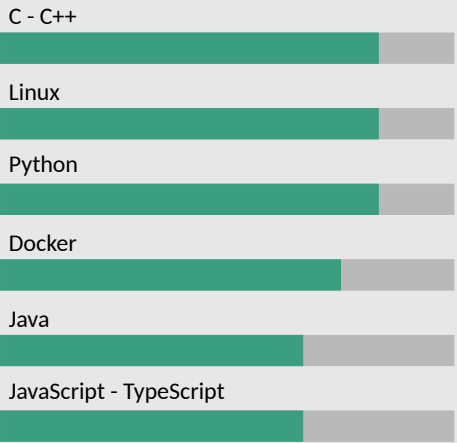
Software Engineer

- Santa Maria da Feira, Portugal
- +351 912768196
- joaocosta.dev
- jobs@joaocosta.dev
- www.linkedin.com/in/joaocostaifg
- github.com/JoaoCostaIFG

About me

Software engineer with a Master's degree in Informatics Engineering. Passionate about building software, managing infrastructure, and sharing knowledge through technical writing.

Skills



Language skills

Portuguese	Native
English	C1

Soft skills

problem-solving flexibility communication

Bio

Versatile software engineer with expertise spanning multiple domains. Professional experience in embedded software development and safety-critical systems for the aerospace industry. Also skilled in system administration and CI/CD.

Continuously expanding technical skills through personal projects covering the full technology stack, including hardware and electronics. Active contributor to open-source software.

Outside of work, I enjoy DIY projects, reading, video games, paintball, and cycling.

Education

2021 - 2023	Master in Informatics Engineering Final grade 18/20	Faculty of Engineering of the University of Porto (FEUP)
2018 - 2021	Bachelor of Science Final grade 18/20	Faculty of Engineering of the University of Porto (FEUP)

Work experience

2024→present	Embedded Software Engineer Developing bare-metal and Linux drivers for DSI controllers and MIPI CD-PHYs. Building internal tooling and CI/CD pipelines to enhance team productivity and streamline software development workflows, including AI based automations. Tech: C, Python, Bash, JavaScript	Synopsys
2024	Backend Software Engineer Contributed to the development of a security-focused product. Led migration of legacy backend services, implemented new API endpoints, and performed database refactoring. Additionally contributed to frontend development in React. Position ended due to company-wide layoffs. Tech: Scala, Java, React	Codacy
2023→2024	Software Engineer Independently designed and developed real-time monitoring systems for metro tunnel construction in Porto. Implemented data acquisition pipelines for sensor networks to ensure worker safety and protect historical infrastructure. Concurrent part-time role alongside position at Critical Software. Tech: Python, C#	Civil Engineering Department of FEUP
2022→2024	Embedded Software Engineer Performed verification and validation of real-time operating systems for aerospace applications. Authored requirements, test cases, and test procedures in compliance with DO-178C standards. Transitioned from Master's thesis internship to full-time role. Tech: C, Python, Arm ASM	Critical Software
2020→2021	Web Game Developer Ported a collection of educational Flash games to the Unity game engine as part of a small development team. The games are used in an educational suite for elementary school children. Part-time position held concurrently with university studies. Tech: Unity, C#	LusoInfo



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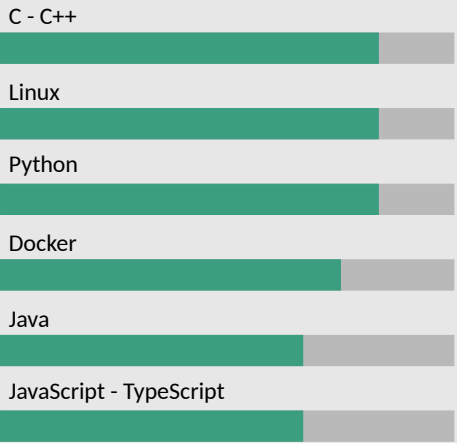
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Awards

- 2023 2nd place in AI for Banking Competition by Millennium BCP
- 2018 Merit-based scholarship awarded by Amorim Cork Composites
- 2018 Top informatics grade in high school graduating class

Projects

- 2026 **joaocosta.dev** [Project's GitHub page](#)
Personal technical blog covering software development and technology topics. Built as a custom Next.js application, evolved through multiple iterations using different technologies.
- 2023 **Octree implementation for SLAM** [Project's GitHub page](#)
High-performance Octree implementation in C++ for SLAM applications in an autonomous submarine. Developed for the university's robotics research institute. Optimized existing implementations to improve computational performance while maintaining spatial efficiency and accuracy.
- 2022 **TraSMAPy** [Project's GitHub page](#)
Python API for the SUMO traffic simulator designed to enable researchers with limited programming experience to build road traffic simulations. Abstracts low-level simulator concepts and provides intuitive interfaces for agent-based modeling.
- 2022 **µKernel** [Project's GitHub page](#)
Real-time micro-kernel for Arduino UNO featuring POSIX-style tasks, mutexes, EDF scheduling, task sleeping, and assertions. Led implementation and research efforts for the project.
- 2021 **Dry Beans classification** [Project's GitHub page](#)
Machine learning model using unsupervised learning to classify beans across seven morphologically similar classes. Contributed to data analysis, preprocessing, algorithm implementation, and comparative evaluation of results.
- 2020-2021 **MAWW** [Project's GitHub page](#)
Linux utility enabling animated desktop backgrounds on X11, compatible with any compositor. Published as an [Arch Linux package](#) on the AUR.

Publications

- 2023 **Verification of real-time operating system for DO-178C compliance**
Master's thesis analyzing and optimizing verification processes for aviation real-time operating systems at Critical Software. Delivered internal tools and process improvements that reduced development time and costs.